

Reading, Not Reasoning: A Causal Faithfulness Audit of Chain-of-Thought in Distilled Chart-and-Table VLMs

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Abstract

Many multimodal systems compress agentic visual reasoning into a single tool-free VLM forward pass via chain-of-thought supervised fine-tuning. We ask what such distillation actually transfers. Our causal claims concern the *emitted written chain*, not hidden-state or latent computation that never surfaces as text. Across chart, table, natural-image counting, and rendered financial QA, CoT SFT reliably improves accuracy—often strongly—yet a causal faithfulness audit shows that the emitted written chain is not load-bearing: corrupting an on-path numeric intermediate does not produce a reliable excess flip over no-target controls, including a targeted same-shape local numeric control, and the model almost never follows the injected value, instead snapping back to the answer readable from the original evidence. The information source behind each gain is task-dependent: the answer remains recoverable from the visible chart or table and the model snaps back to it, is reconstructed from redundant enumerations in the rationale, or is copied from the chain’s own conclusion. Even a shortcut-removing curriculum with gold-program-verified operands raises accuracy substantially without installing load-bearing intermediate computation. Accuracy-only internalization claims conflate reading with reasoning; causal probes over emitted rationales can separate the two.

1 Introduction

Agentic VLM systems—iterative frame retrieval, self-reflection, critic-orchestrated re-reading, visual tool use—often improve accuracy, but their inference-time tools are expensive. A natural compression step is to convert an expert trajectory into a written rationale or latent trace and fine-tune a VLM to execute one forward pass at test time (Adhikari and Lapata, 2026; Carbune et al., 2024; Hu et al., 2024; Tao et al., 2026; Wei et al., 2026). Accuracy improvements from such distillation are routinely reported, but a prior question is left unasked: when distillation lifts accuracy, does the student’s emitted CoT carry the causal work, or does it merely record, paraphrase, or reveal an answer found elsewhere?

We address this question directly. We assess causal faithfulness by corrupting emitted numeric intermediates and comparing against no-target controls that range from harsh order disruption to gentler same-shape local numeric perturbation, plus a targeted semantic counterfactual diagnostic (Lanham et al., 2023). The causal claim is about the emitted written chain; the probes do not rule out hidden-state or latent computation that never surfaces in the rationale.

*Draft of 2026-06-26. All numbers regenerate from the project’s append-only result store; see §8.

Contributions.

1. We show that CoT SFT reliably and substantially improves accuracy on chart, table, natural-image counting, and rendered financial QA—establishing that there is a real accuracy signal for the faithfulness audit to interrogate.
2. We demonstrate that accuracy gains and causal faithfulness decouple: a causal probe battery across four task types finds that the emitted rationale is not load-bearing, even when accuracy improvements are large and statistically robust.
3. We characterize the task-dependent information source behind each gain—re-reading visible evidence, exploiting redundant enumerations, or copying the chain’s conclusion—and show these are distinguishable by the probe design.
4. We provide a pre-specified test showing that shortcut-removing curricula with gold-program-verified operands raise accuracy without installing load-bearing intermediate computation, ruling out a straightforward training-data explanation for the faithfulness null.

2 Related Work

Internalizing visual reasoning. Visual Program Distillation (Hu et al., 2024), chart-specific rationale transfer (Carbune et al., 2024; Chen et al., 2025), latent trajectory learning (Adhikari and Lapata, 2026; Li et al., 2026), and region-to-image or local-cue distillation (Tao et al., 2026; Wei et al., 2026) all aim to move work that previously required tools, crops, or expert traces into a cheaper model or inference path. These papers mainly validate the transfer by answer accuracy, sometimes with attention or qualitative trace analysis. Our question is narrower: after a no-tool VLM emits a discrete rationale, is the emitted intermediate value an answer-carrying state, or is it an artifact of a successful read?

Chain-of-thought and its faithfulness. Chain-of-thought prompting (Wei et al., 2022), its zero-shot trigger (Kojima et al., 2022), and self-consistency decoding (Wang et al., 2023) made written intermediate steps a default interface for reasoning, which in turn raised the question of whether those steps are faithful to the computation that produces the answer. A broader explanation-evaluation literature warns that a plausible rationale need not be the true cause of a prediction (Jacovi and Goldberg, 2020), that rationale datasets and metrics measure heterogeneous properties (Wiegreffe and Marasović, 2021), and that rationalized models must be benchmarked against the evidence they actually use (DeYoung et al., 2020). Within text LLMs specifically, rationale scaffolds can be made more explicitly faithful (Lyu et al., 2023), plausible rationales can omit true decision drivers (Turpin et al., 2023), larger models can produce less answer-dependent CoT (Lanham et al., 2023), and filler tokens can support hidden computation independent of human-readable steps (Pfau et al., 2024). Two critiques matter for our design: unlearning or deletion can measure contextual robustness rather than load-bearing computation (Tutek et al., 2025), and multiple-choice option bias can make some faithfulness metrics look like disguised accuracy (Bentham et al., 2024). In multimodal settings, recent audits target different objects: visual-thinking agents may ignore visual evidence (Liu et al., 2025b), CodeV perturbs tool-output images rather than internalized CoT tokens (Hou et al., 2026), and latent imagination work studies continuous intermediate states (Li et al., 2026).

Chart/table QA and tool/program distillation. Chart and table QA are useful because the same task mixes visual extraction and numerical reasoning. Benchmarks from synthetic figures

(Kafle et al., 2018; Kahou et al., 2018; Methani et al., 2020) to realistic charts (Masry et al., 2022; Tang et al., 2025; Wang et al., 2024), semi-structured math tables (Lu et al., 2023), and hybrid financial tables (Chen et al., 2021; Zhu et al., 2021), together with chart-noise stress tests (Shin et al., 2025) and perception-bottleneck analyses (Liu et al., 2025a), all point to a persistent ambiguity: an accuracy improvement can come from reading values better, computing with them better, or both. A parallel line tries to install the computation explicitly, either by emitting executable programs at inference (Chen et al., 2023; Gao et al., 2023), learning to call tools (Schick et al., 2023), or distilling tool traces and step-by-step rationales into a cheaper model (Hsieh et al., 2023; Hu et al., 2024). Our experiments exploit the reading/computing ambiguity instead of hiding it: by masking images, falsifying on-path operands, and comparing chart/table with natural-image counting, we ask which representation the answer actually reads from at the final step.

Positioning. Table 1 places this work against neighboring lines along four axes: the object audited, the intervention used, the model setting, and the conclusion the design licenses. The distinction is not that the CORRUPT intervention is novel—it is a standard counterfactual edit (Lanham et al., 2023)—but that the audited object is the discrete CoT of a *distilled, single-forward* VLM whose visual evidence is fixed in the input, evaluated with controls that separate re-reading, redundancy, and conclusion copying. That combination lets us ask not just whether the chain is faithful, but which fixed source the answer reads from when it is not.

3 Method: Causal Faithfulness Audit

This section is not only a list of datasets, models, and metrics: it defines the audit that is one of the paper’s contributions. We specify how a targeted counterfactual edit (CORRUPT), a battery of no-target controls (SHUFFLE, LOCAL-NUM, and others), and a snap/follow/other readout together decide whether an emitted written chain is *load-bearing*—whether the final answer causally tracks an edited intermediate, or instead reads from a source still available at answer time.

Definitions. We fix three terms used throughout. A *load-bearing written chain* is an emitted numeric intermediate whose value causally determines the final answer: under a counterfactual edit to that token, a load-bearing chain forces the answer to track the edited value. *Reading* is the complementary case in which the final answer is recovered from a source still available at answer time—visible pixels, a redundant enumeration elsewhere in the rationale, or the chain’s own conclusion line—so that editing an upstream intermediate leaves the answer unchanged. *Latent reasoning* is any computation that does not surface as an edited written token, including arithmetic that the model may perform while re-reading the image. Our probes intervene only on emitted written tokens; they can show that a written intermediate is not load-bearing, but they cannot certify the absence of latent reasoning. “Reading, not reasoning” is therefore a claim about the emitted chain, not about the model’s internal state.

Tasks. *ChartQA* (Masry et al., 2022) (400 held-out open numeric questions, image-disjoint from SFT train images), *TabMWP* (Lu et al., 2023) (400 rendered-table, free-response numeric cases), TallyQA-complex natural-image counting (400 open integer questions), and FinQA rendered tables for the pre-specified curriculum probe. All four tasks fix the visual evidence in the input, making the causal question well-posed: does the written chain carry the computation that produces any SFT accuracy gain?

Table 1: Field positioning. Where a causal CoT audit of distilled chart/table VLMs sits relative to neighboring lines of work. “Object audited” is the representation perturbed; “conclusion licensed” is what the design can actually claim.

Line of work	Object audited	Intervention	Model setting	Conclusion licensed
Text CoT faithfulness (Bentham et al., 2024; Lanham et al., 2023; Pfau et al., 2024; Turpin et al., 2023; Tutek et al., 2025)	Written CoT tokens	Truncate / paraphrase / corrupt / unlearn	Prompted text LLM	Whether text CoT is answer-bearing
Rationale / explanation eval. (DeYoung et al., 2020; Jacovi and Goldberg, 2020; Wiegrefe and Marasović, 2021)	Highlighted rationale / evidence	Erasure, comprehensiveness, sufficiency	Classifier + rationale	Whether a rationale matches used evidence
Visual-agent faithfulness (Hou et al., 2026; Liu et al., 2025b)	Tool-output images / agent traces	Perturb tool outputs / RL reward	Tool-using VLM agent	Whether agents use visual tool results
Tool / program distillation (Chen et al., 2023; Gao et al., 2023; Hsieh et al., 2023; Hu et al., 2024; Schick et al., 2023)	Program / tool trace	Train-time supervision; run program	Distilled / program-aided model	That accuracy transfers (faithfulness untested)
Latent-state reasoning (Adhikari and Lapata, 2026; Li et al., 2026)	Continuous latent states	Probe / edit embeddings	Latent-trace VLM	Whether latent states carry reasoning
Chart/table QA benchmarks (Chen et al., 2021; Kafle et al., 2018; Lu et al., 2023; Masry et al., 2022; Methani et al., 2020)	Final answer only	None (accuracy scoring)	Any VLM/LLM	Aggregate accuracy on visual+numeric QA
This work	Emitted CoT of distilled single-forward VLM	Targeted numeric corrupt + no-target controls + mask	Distilled VLM, fixed visual input	Which fixed source the answer reads from

Models. The main SFT and Qwen-family faithfulness cells use Qwen3-VL 8B and 32B (dense) (Bai et al., 2025). For cross-family external validity, we also run the same ChartQA present-image probe on InternVL3.5-8B (Wang et al., 2025), a non-Qwen VLM served through an OpenAI-compatible local endpoint. All models are served locally.

SFT pipeline. For ChartQA and TabMWP, a Qwen3-VL-32B teacher generates step-by-step rationales on training images; we keep only traces whose final answer matches the gold answer under the numeric grader below. This yields 150 ChartQA traces and 160 TabMWP traces. The 8B students are LoRA-SFT models with the vision tower frozen, rank 16, $\alpha=32$, dropout .05, bf16, learning rate 10^{-4} , per-device batch size 1, gradient accumulation 8, cosine schedule, and prompt tokens masked to -100 so supervision applies only to the assistant rationale and ANSWER line. The ChartQA/TabMWP 8B runs use two epochs, corresponding to 38 and 40 optimizer updates. The 32B students use the same LoRA target modules under NF4 QLoRA and run five epochs with greedy held-out evaluation after each epoch; the selected checkpoints are epoch 1 for ChartQA and epoch

4 for TabMWP. Dense 8B controls use full-SFT checkpoints with the vision tower, embeddings, and first three language layers frozen, leaving about 79.7% of parameters trainable. The FinQA curriculum arms use the same LoRA recipe for three epochs: 180/180 gold-program strict/vanilla traces, and 173/173 answer-conditioned text-teacher strict/vanilla traces. All answer generation and probe continuations use greedy decoding (`do_sample=False`; answer max-new 320–384 tokens, continuation max-new 64 tokens).

Probe construction. For each task/model cell, we first ask the evaluated model to solve the original visual question with the image/table visible and the prompt `Solve step by step, end with 'ANSWER: <final>'`. The text before the first `ANSWER:` is the emitted CoT. A case is probe-eligible only if this CoT is nonempty and the extracted final answer is correct. Empty rationales, abnormal outputs, failed final-answer parses, and wrong final answers are not edited; they re-enter the analysis only through the selection-flow and sensitivity bounds below. Eligible CoTs with no corruptible numeric token remain in the flow but are excluded from the CORRUPT denominator and treated as corrupt-unusable in sensitivity bounds. The generic chart/table battery then edits that model-owned CoT with a fixed edit seed `Random(0)`: CORRUPT samples one numeric token from the CoT and replaces value v by $2v + 7$ (integers remain integers; decimals are rendered to one decimal), while SHUFFLE shuffles sentence order without injecting a target value. Additional controls paraphrase the CoT with an exact number-multiset guard, replace tokens with length-matched filler, truncate early sentences, or delete the last number-bearing steps. P0-4 adds two ChartQA 8B replications. The gentler LOCAL-NUM control excludes the CORRUPT span, replaces a different numeric token with a same-shape value, preserves sentence order and local syntax, and avoids the gold/base/injected values when possible. The SEMANTIC-CF diagnostic sends the probe-eligible CoT prefix to a configured LLM rewrite endpoint, changes non-final numeric reasoning with dependent non-final arithmetic made locally coherent, and validates that the protected final conclusion segment is preserved rather than directly overwritten. The FinQA targeted probe is stricter: it uses the gold program to choose a table operand that is non-constant, appears in the student’s CoT, and whose replacement by $2v + 7$ changes the gold-program final answer; it then tests both operand-only corruption and a consistently propagated corrupted chain. Each edited CoT is force-continued with the prompt “Reasoning so far: ... Given ONLY the reasoning above, state the final answer now as `ANSWER: <final>`”. The masked condition drops the image/table from this force-continue call while keeping the same edited text.

Grading and readout. Open numeric answers are graded in the task’s original answer units. We strip commas and percent signs from the gold answer, extract all numeric strings from the prediction, and mark a match if any predicted number is within $0.05|g| + 10^{-6}$ of gold value g ; nonnumeric gold answers fall back to normalized substring matching. Rounding and redundant units are therefore tolerated only through the numeric value; symbolic equivalents that do not surface a matching number, empty outputs, and parse failures are incorrect. “Flip” means the forced final answer no longer matches the unedited model answer. For CORRUPT and FinQA targeted probes, the three-way readout is applied in priority order: snap if the answer matches the gold answer, follow if it does not snap but matches the injected wrong value (or injected gold-program final), and other otherwise. Boundary examples under 5% tolerance are: gold 100, injected 207, answer “about 96” → snap; answer “207” → follow; answer “200” → follow; answer containing both 100 and 207 → snap by precedence; and a SHUFFLE answer has no injected target, so off-gold responses are counted as other rather than follow.

Original CoT: read value a , read value b , compute $a - b = c$, answer c . CORRUPT: read value a , read value b , compute $a - b = \tilde{c}$, answer ? SHUFFLE: same rationale tokens, shuffled order, answer ? LOCAL-NUM: same rationale order, different local number perturbed, answer ? Readout: snap to true c / follow injected $\tilde{c}$ / other.

Figure 1: Probe schematic. CORRUPT tests whether a falsified written intermediate is load-bearing; SHUFFLE is a harsh no-target order control, while LOCAL-NUM is a gentler same-shape local numeric control.

Statistics. McNemar exact tests for paired SFT deltas, using only discordant pairs; Holm correction across the four headline ChartQA/TabMWP cells leaves all tests below .01. At $n=400$, minimum detectable net is about 8–9 points, and the observed paired SFT gains have 66 and 46 discordant pairs at 8B/32B. For the faithfulness probe, the load-bearing claim is the one-sided test $\text{flip}_{\text{CORRUPT}} > \text{flip}_{\text{SHUFFLE}}$, Holm-corrected across the eight Qwen chart/table cells, with Wilson intervals on snap/follow and bootstrap intervals on F (§5).

Robustness controls. We do not claim the CORRUPT intervention itself is new; it is useful only as part of a controlled battery. To address the “contextual faithfulness” concern (Tutek et al., 2025), we compare CORRUPT against SHUFFLE, paraphrase, filler-token, deletion, truncation, image-mask, a targeted LOCAL-NUM replication, and a snap/follow/other readout that asks whether the final answer follows the injected value. To address disguised-accuracy and option-bias concerns (Bentham et al., 2024), the main faithfulness substrates are free-response numeric QA rather than multiple choice. To address the concern that force-continue is a two-pass re-prompt rather than an on-stream causal intervention, we additionally validate with an IN-PLACE paradigm that corrupts a token inside the model’s own greedy generation stream and lets it continue (Result II, finding 7). To avoid reporting a probe artifact as mechanism, we treat a positive corruption flip as insufficient by itself; the headline evidence is convergence across controls and regimes.

4 Result I: CoT SFT Improves Accuracy

On the chart and table tasks, we run the actual distillation. A Qwen3-VL teacher generates step-by-step rationales; answer-consistency filtering keeps high-quality traces; LoRA/QLoRA SFT trains a single-forward student. All reported evaluation sets are image-disjoint from training and use open numeric grading. These gains are best read as improved single-pass readout unless the causal probe shows that the emitted chain itself carries the answer.

- **ChartQA**, $n=400$: 8B .713 $\rightarrow$ **.778**, net +6.5 points, CI [+2.5, +10.5], McNemar $p=0.0021$; 32B .725 $\rightarrow$ **.795**, net +7.0 points, CI [+3.8, +10.3], McNemar $p=6.9 \times 10^{-5}$. The original 32B $n=60$ pilot had only four discordant pairs; the expanded evaluation has 37 gains vs. 9 losses.
- **TabMWP**, $n=400$: 8B .865 $\rightarrow$ **.963**, net +9.8 points, McNemar $p=2.9 \times 10^{-9}$; 32B .835 $\rightarrow$ **.973**, net +13.8 points, McNemar $p=2.1 \times 10^{-12}$.

Thus the paper is not a null-result critique of SFT. The accuracy gains are real, paired, and adequately powered on static chart/table substrates. The remaining question is what mechanism those gains bought.

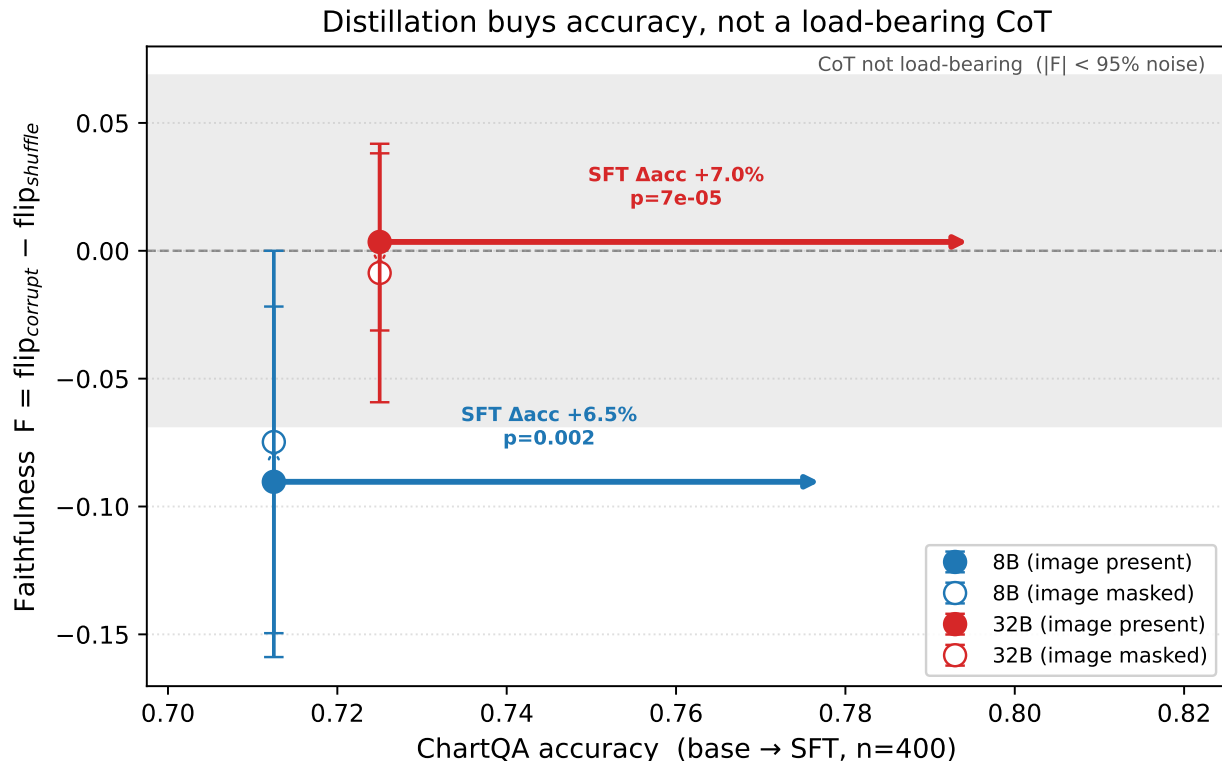


Figure 2: Accuracy $\perp$ written-chain faithfulness (chart/table = pixels-source regime). CoT distillation lifts ChartQA accuracy by +6.5–7.0% (McNemar $p \leq .002$; horizontal arrows), yet the causal CoT metric stays on the $F \approx 0$ band ($F = \text{flip}_{\text{CORRUPT}} - \text{flip}_{\text{SHUFFLE}}$): corrupting a numeric intermediate flips the answer no more than shuffling the CoT. This is a conservative harsh-control criterion: SHUFFLE disrupts order without injecting any target value, so $F \leq 0$ means the falsified intermediate contributes no extra flip beyond order/format noise. A targeted ChartQA 8B LOCAL-NUM replication below checks the same claim under a gentler no-target perturbation. Hollow markers show image-masked controls. F is measured on the same SFT student used for the corresponding probe; the tabular intervals and corrected tests in Table 2 are regenerated by `scripts/faithfulness_stats.py`.

Dense 8B control. To rule out a low-rank-capacity explanation, we repeat the 8B arms with dense/full-SFT controls (vision tower, embeddings, and the first three language layers frozen; about 79.7% trainable parameters for ChartQA/TabMWP). The accuracy deltas remain positive: ChartQA .713 $\rightarrow$.763 (net +5.0 points, 39 gains/19 losses, $p=0.0126$) and TabMWP .865 $\rightarrow$.968 (net +10.3 points, 42 gains/1 loss, $p=1.1 \times 10^{-9}$). Available dense probes keep the LoRA signature: ChartQA’s completed six-intervention battery has present-image CORRUPT/SHUFFLE = .100/.259, snap .917 and follow .030; TabMWP’s completed core battery has present-table CORRUPT/SHUFFLE = .049/.161, snap .971 and follow .016; FinQA dense targeted arms keep operand-follow near zero while consistent-conclusion follow remains .947–.982. The remaining dense-control gap is the full six-intervention TabMWP present/masked battery, so the dense control is evidence against the LoRA-capacity confound but not yet the final multi-cell table.

Table 2: Causal CoT-intermediate probe on chart/table SFT cells. Flow is raw $n=400$ to probe-eligible correct+CoT cases, with CORRUPT/SHUFFLE denominators in parentheses. $F = \text{flip}_{\text{CORRUPT}} - \text{flip}_{\text{SHUFFLE}}$; p_H is Holm-adjusted for the one-sided load-bearing test. Snap/follow are rates after CORRUPT with Wilson 95% CIs.

Dataset/model	Cond.	raw→probe	CORRUPT/SHUFFLE	F [95% CI]	p_H	snap [95% CI]	follow [95% CI]
ChartQA 8B	present	400→321 (321/321)	.212/.302	−.090 [−.159, −.022]	1.000	.816 [.770, .855]	.031 [.017, .056]
ChartQA 8B	masked	400→321 (321/321)	.333/.408	−.075 [−.150, +.000]	1.000	.710 [.658, .757]	.100 [.072, .137]
ChartQA 32B	present	400→318 (316/318)	.051/.047	+ .003 [−.031, +.038]	1.000	.981 [.959, .991]	.016 [.007, .036]
ChartQA 32B	masked	400→318 (316/318)	.120/.129	−.009 [−.059, +.042]	1.000	.908 [.871, .935]	.057 [.036, .088]
TabMWP 8B	present	400→385 (385/385)	.034/.145	−.112 [−.151, −.073]	1.000	.971 [.950, .984]	.008 [.003, .023]
TabMWP 8B	masked	400→385 (385/385)	.278/.299	−.021 [−.086, +.044]	1.000	.722 [.675, .764]	.088 [.064, .121]
TabMWP 32B	present	400→389 (389/389)	.033/.072	−.039 [−.069, −.008]	1.000	.969 [.947, .982]	.015 [.007, .033]
TabMWP 32B	masked	400→389 (389/389)	.075/.103	−.028 [−.069, +.010]	1.000	.928 [.898, .950]	.039 [.024, .063]

Table 3: P0-5 cross-family replication. InternVL3.5-8B is a non-Qwen VLM evaluated on the same ChartQA present-image probe protocol. This row is external-validity evidence and is reported separately from the Qwen-family SFT cells in Table 2.

Model family	raw→probe	base acc	CORRUPT/SHUFFLE	F [95% CI]	p	snap [95% CI]	follow [95% CI]
InternVL3.5-8B	400→302 (302/302)	302/400 (.755)	.070/.096	−.026 [−.053, −.003]	.994	.990 [.971, .997]	.000 [.000, .013]

5 Result II: The Causal Probe—the Written Chain Is Not Load-Bearing

For every correctly answered case with an emitted CoT, we intervene on the rationale and force-continue to a final answer. If a numeric intermediate carries the computation, CORRUPT should flip more often than no-target controls, and the final answer should follow the injected value rather than snap back to the original truth. Table 2 reports the high-power Qwen-family ChartQA and TabMWP CORRUPT-vs.-SHUFFLE cells with uncertainty; this family treats SHUFFLE as a deliberately harsh order/format control, not as the only clean control. The confirmatory family is the eight Qwen chart/table CORRUPT-vs.-SHUFFLE cells; the load-bearing alternative is one-sided, $\text{flip}_{\text{CORRUPT}} > \text{flip}_{\text{SHUFFLE}}$, with Holm correction across that family. Table 3 separately reports the P0-5 non-Qwen replication and is not included in that Holm family. Snap/follow intervals are Wilson 95% intervals. The legacy headline battery stores per-case CORRUPT answers and aggregate SHUFFLE margins, so F intervals use an aggregate binomial bootstrap and the test is an exact randomization conditional on the recorded margins; newer battery files with `details[].answers` use paired case bootstrap and paired exact sign-randomization under the same script. The P0-4 LOCAL-NUM and SEMANTIC-CF ChartQA 8B replications are reported separately in the text and in `faithfulness_stats.md`.

Seven findings follow.

1. **ChartQA and TabMWP agree.** In every present-image/table cell, $\text{CORRUPT} \leq \text{SHUFFLE}$ within the 95% interval, and none of the Holm-corrected load-bearing tests is close to significant. The model usually snaps back to the true value and rarely follows the injected value. TabMWP is the cleanest case: despite large SFT gains, follow is .008 [95% CI .003, .023] and .015 [.007, .033] at 8B/32B.
2. **The causal readout reproduces off Qwen.** InternVL3.5-8B answers 302/400 ChartQA cases correctly with an emitted CoT and shows the same non-load-bearing signature: $\text{CORRUPT/SHUFFLE} = .070/.096$, $F = -.026$ [95% CI $-.053, -.003$], and follow is 0/302

Table 4: Selection flow and answer-eval strata. Answer-eval strata are stable-correct / SFT-gain / SFT-loss / both-wrong on the paired $n=400$ accuracy run. Probe-kept strata are the same case IDs after the separate CoT-generation eligibility filter. “No usable corrupt” is raw cases without a usable CORRUPT readout, including non-eligible cases and any eligible CoT with no corruptible numeric token. The final columns report present-condition CORRUPT follow rates in the two strata most relevant to the main claim. Full masked rows, loss/wrong strata, and sensitivity bounds are in `faithfulness_stats.md`.

Dataset/model	answer-eval strata	probe-kept strata	SFT correct	probe eligible	no usable corrupt	stable follow	gain follow
ChartQA 8B	265/46/20/69	262/39/10/10	311	321	79	9/262 (.034)	0/39 (.000)
ChartQA 32B	281/37/9/73	281/37/0/0	318	318	84	5/281 (.018)	0/37 (.000)
TabMWP 8B	345/40/1/14	345/40/0/0	385	385	15	3/345 (.009)	0/40 (.000)
TabMWP 32B	332/57/2/9	332/57/0/0	389	389	11	6/332 (.018)	0/57 (.000)

[95% CI .000,.013] while snap is .990. This row does not prove Qwen-style SFT transfers to InternVL; it addresses the narrower reviewer concern that low follow/high snap might be a Qwen3-VL-specific causal-probe behavior.

3. **The gain subset itself is reading.** We join the probe-kept cases to the paired SFT evaluation labels (Table 4). On probe-kept cases newly solved by SFT, present-condition follow is zero in every headline cell: ChartQA 8B 0/39 [95% CI .000,.090], ChartQA 32B 0/37 [.000,.094], TabMWP 8B 0/40 [.000,.088], and TabMWP 32B 0/57 [.000,.063]. FinQA is not reported as a matched gain subset because the targeted battery is keyed by gold-program operand eligibility, not by base-wrong $\rightarrow$ SFT-right eval partitions; its operand-follow intervals are nevertheless tight: 0/172 at 8B [0,.022] and 0/175 at 32B [0,.021]. The new correct answers are not being computed from the falsified written arithmetic.
4. **Selection does not carry the result.** The estimand is intentionally conditional on an emitted, correct written rationale: an incorrect or missing rationale cannot be asked whether its intermediate value was load-bearing for a correct answer. Still, the selection flow is visible. TabMWP keeps almost every SFT-correct case (385/385 at 8B, 389/389 at 32B), and the gain strata have zero present-condition follows. ChartQA is noisier: 8B has 10 SFT-correct answer-eval cases missing from probe eligibility and 20 probe-eligible cases that were not SFT-correct in the separate answer-eval pass, so we report them rather than silently dropping them. A conservative sensitivity bound that assigns every unobserved raw case to follow gives a loose all-raw upper bound on follow (.223 for ChartQA present cells), but this adversarial bound contradicts the directly observed SFT-correct subset: even if every SFT-correct ChartQA 8B case missing from the probe followed the injected value, SFT-correct present-condition follow would be at most $19/311=.061$; ChartQA 32B is at most $7/318=.022$, and TabMWP remains below .016. The conclusion therefore does not rely on choosing only stable-correct cases.
5. **Masking diagnoses the information source.** On tables, masking the image raises flip and “other” errors sharply, especially for 8B, but follow remains small. The masked contrast is the key diagnostic: if the model were obeying the corrupted chain, removing the image should leave follow unchanged or higher; instead follow stays low while “other” rises, which means the model is neither faithfully using the bad chain nor injecting the falsified value, but losing the source it actually relied on. We therefore describe the visible-input mechanism conservatively: the answer remains recoverable from the visible chart or table and the model snaps back to it. We do not run targeted cell/bar/region editing, so we do not claim a literal re-reading saccade; what the data license is recoverability of the answer from fixed visual evidence, not the obeyed written intermediate.

6. **Other interventions point the same way, with one diagnostic caveat.** Paraphrase behaves like CORRUPT (e.g., 32B ChartQA present .053 vs. .051), while filler/truncation/deletion and SHUFFLE mainly test format and token-budget sensitivity. Because SHUFFLE injects no target value, its paper-facing readout is flip/snap/other rather than follow. Existing summaries show this is a harsh control rather than an injected-value test: present-image/table SHUFFLE accuracy-after is .735/.987 on ChartQA 8B/32B and .857/.936 on TabMWP 8B/32B. A gentler P0-4 LOCAL-NUM replication on ChartQA 8B gives the same qualitative null under paired analysis: CORRUPT/LOCAL-NUM = .235/.224 on the 294 cases with both continuations, $F_{\text{local}} = +.010$ [95% CI $-.034, +.054$], paired exact $p = .385$, with low CORRUPT follow 13/321=.040 [95% CI .024,.068]. A separate SEMANTIC-CF diagnostic successfully rewrites 288/321 eligible CoTs; it is a stronger whole-prefix intervention, and indeed flips more often than single-token corruption (CORRUPT/SEMANTIC-CF = .236/.434, $F_{\text{semantic}} = -.198$ [95% CI $-.257, -.139$], paired exact one-sided $p = 1.000$ for CORRUPT > SEMANTIC-CF). We therefore do not use SEMANTIC-CF as a gentle control; it shows that broad semantic rewriting can perturb answers, while the targeted numeric CORRUPT intervention still fails the load-bearing and follow tests. The conclusion is carried by the whole battery, not by a single corruption statistic.
7. **A generation-time intervention strengthens, not weakens, the conclusion—with one honest correction.** The force-continue probe is a two-pass re-prompt (a new prompt with an added instruction, “Given ONLY the reasoning above, state ANSWER:”), so a skeptic can argue that low present-image follow merely reflects the model re-reading the image on a second forward pass rather than the written chain being non-load-bearing. We therefore add an IN-PLACE paradigm that stays inside the model’s own autoregressive stream: we re-feed the model’s *own* greedy tokens up to the boundary before a selected numeric token, substitute the corrupted value, and let the model *continue generating* its own chain and answer—no re-prompt, no added instruction, and the conclusion line is never supplied. A paradigm-validity control that re-feeds the *true* value and continues reproduces the model’s base answer on 287/307 cases (.935), certifying that the re-encoded continuation tracks the original stream. We pre-registered the threshold before running (§8). The outcome is not a free pass: on the same ChartQA 8B student, IN-PLACE CORRUPT flips more than the two-pass probe (.293 [95% CI .245,.346] vs. .098) and follow rises to .094 [95% CI .067,.132] from .036, so the re-prompt paradigm *under*-estimates the chain’s causal influence and we report that rather than claiming clean convergence. Yet the core reading survives the stricter test: follow remains low and snap still dominates (.733 [95% CI .681,.779]), and most of the extra movement is into “other” (53/307, .173)—stream-level corruption derails generation more than re-prompting but rarely makes the model adopt the injected value as its answer. A no-visual variant (image absent throughout) keeps follow at 0/23 but is low-powered (base accuracy .07 without the chart) and its clean control degrades to .52, so we treat it only as a secondary diagnostic consistent with the masked force-continue finding. The full cross-paradigm table is in `faithfulness_stats.md`.

6 Result III: Task-Dependent Information Sources and a Constructive Test

The chart/table probe (§5) isolates the *pixels* regime, where the answer reads from fixed visual evidence. The two substrates below are complementary diagnostics, not the same SFT-mechanism evidence restated: each tests a *different* readable source. Natural-image counting is an external regime check on *base* models that probes the *redundant-text* source, and the FinQA curriculum

probes the *conclusion-line* source. We keep them separate rather than aggregating them into one causal claim.

Natural-image counting (external regime check; redundant-text source). To test whether the chart/table result is a chart artifact, we run a pre-specified TallyQA-complex counting probe on *base* Qwen3-VL-8B/32B, with no chart adapter. We treat this as an external regime check on the base model, not as SFT-internalization evidence on par with the ChartQA/TabMWP cells: there is no distillation step here, so it tests only whether the non-load-bearing readout generalizes to a different visual regime, not whether an SFT gain was internalized. This is a genuinely perception-bottlenecked natural-image regime (free-form accuracy .415/.448). Yet the emitted CoT’s numeric content is again not load-bearing: across scales, follow is 0/345 with the image present and 1/345 when masked. Even when CORRUPT hits the count token itself, follow is 0/103 present and 1/103 masked. The model reconstructs the answer from redundant enumeration in the text, so masking the image is almost inert. This is the same reading-not-reasoning pattern with a different readable source: not the image, but the redundant chain.

Shortcut-removing curriculum (SFT; conclusion-line source). As a constructive counter-test, we train on rendered FinQA tables filtered to require multi-step, multi-cell arithmetic with a gold-program-verified flippable operand. The curriculum uses a deterministic gold-program teacher and a matched vanilla control; a text-teacher arm repeats the test with fluent answer-conditioned rationales. Accuracy rises substantially (8B: .080 base $\rightarrow$.582 vanilla $\rightarrow$.674 curriculum; 32B: .123 $\rightarrow$.659 $\rightarrow$.682). This probe yields *two distinct* readings that we deliberately do not merge. First, *operand-follow is near zero*: changing a verified on-path *input* operand yields follow 0/172 for the 8B curriculum and 0/175 for the 32B curriculum, so the emitted operands are not the load-bearing source. Second, *conclusion-line follow is high*: when the chain’s final conclusion is consistently rewritten to the recomputed value, the model follows it in 96–97% of cases, so the answer reads from the conclusion line. The observable mechanism is conclusion copying, not recomputation from the emitted operands under this probe; we do *not* extend this to the broader claim that no recomputation occurs—the model may perform arithmetic latently, and this probe cannot see it. We only establish that the written operands are not the load-bearing source. Thus even a shortcut-removing curriculum can improve accuracy without installing a load-bearing written chain.

7 Discussion and Limitations

Implications. (i) Claims of “internalizing agentic reasoning” need causal audits. Significant accuracy gains are compatible with recovering the answer from still-visible evidence, reconstructing from redundant rationale text, or copying the rationale’s conclusion. (ii) The load-bearing locus is task-dependent: answer-time evidence can live in pixels, in redundant text, or in a conclusion line. A single CoT faithfulness score without information-source controls is under-specified.

Deployment audit implications. The probe is meant to be reused, not only to criticize. We read it as a small audit procedure for teams that ship CoT-style explanations on chart/table/figure QA. A minimal checklist: (1) before trusting an emitted rationale, run the targeted CORRUPT-vs.-no-target battery on a held-out slice and report F and the snap/follow split, not accuracy alone; (2) *show* a CoT to end users only when follow is non-trivial and F is positive, i.e. the written chain is load-bearing; (3) when follow is near zero and the answer snaps back, either suppress the chain or label it as a post-hoc summary rather than the decision path; (4) set a faithfulness threshold per

deployment (e.g. a minimum load-bearing F with its CI excluding zero) and treat regressions below it as release blockers; (5) attach a short risk-disclosure note stating which fixed source the answer reads from (pixels, redundant text, or the conclusion line). Two misuse directions are symmetric and both wrong: an unfaithful CoT is *not* a license to withhold explanations entirely, and a fluent but non-load-bearing CoT must *not* be displayed to manufacture a false sense of transparency. The intended users are benchmark authors who report CoT internalization gains, VLM system builders who decide whether to surface rationales, XAI auditors who certify explanation faithfulness, and operators of high-stakes table/chart QA (finance, scientific, and clinical reporting) where a plausible-but-decorative chain is an active hazard. In this framing the contribution is not a critique of CoT SFT claims but a reusable causal-audit practice.

Limitations. All causal claims are about emitted rationales, not a proof that the model lacks latent internal computation. In particular, the re-reading step we infer from the snap behavior may itself involve latent arithmetic over the visible evidence; our probes do not exclude that. What they do establish is narrower and specific: the final answer is not read off the corrupted written intermediate, since editing that token does not move the answer beyond no-target controls. The chart/table and FinQA operands are input-readable by design, which is exactly why these substrates can distinguish re-reading from written-chain use; the scope does not include tasks whose required operands are unavailable in the input. The 32B students use QLoRA/NF4 rather than full bf16 fine-tuning. The main teacher family is Qwen3-VL, although the FinQA text-teacher arm reduces a format-specific teacher concern. Corrupting one numeric token is a sensitivity lower bound; we therefore report it only with shuffle/paraphrase/filler/truncation, targeted local numeric and semantic-counterfactual controls, image masking, snap/follow/other classification, and targeted gold-program operand tests. The LOCAL-NUM and SEMANTIC-CF controls are currently ChartQA 8B replications rather than full eight-cell reruns; SEMANTIC-CF uses a protected-final rewrite cache and is best read as a semantic diagnostic rather than a format-matched local control. Legacy batteries did not persist raw generated CoTs for non-eligible cases, so we cannot retrospectively split no-CoT, wrong-answer, and parse-failure reasons inside the nonprobe mass; instead, Table 4 reports the exact answer-eval and probe-kept strata and `faithfulness_stats.md` gives lower/upper follow bounds that assign the whole unobserved mass adversarially.

8 Conclusion

We set out to compress tool-scaffolded visual reasoning into one forward pass and to verify what moved. The answer is not a blanket dismissal of CoT or of distillation: SFT works, often strongly. The causal result is more specific. In these chart, table, natural-counting, and rendered-FinQA regimes, the emitted answer reads from the easiest available source: pixels, redundant enumerations, or the chain’s final conclusion. Accuracy can move while load-bearing written-chain faithfulness does not. That is the audit lesson behind *Reading, Not Reasoning*.

Reproducibility Statement

All numbers regenerate from a fingerprinted, append-only result store (`results.jsonl`) via the released scripts; SFT, merge, battery, curriculum, and probe scripts are included. ChartQA/TabMWP builders, power table, faithfulness figures, N3 natural-counting probe, and N1 targeted curriculum probe each correspond to one script with fixed seeds or saved artifacts. The generation-time IN-PLACE paradigm (`battery_online_p3.py`) and its convergence threshold were pre-registered before the run

in docs/preregistration_p3_inplace.md; both paradigms in the cross-paradigm comparison run on the same student so the only varied factor is prompt-level versus stream-level intervention.

A Artifact Map

Table 5 gives the minimum artifact map needed to reconstruct each headline table or figure without reading source code first. Paths are relative to the repository root. JSON hashes are 12-character SHA-256 prefixes of the files used in this draft.

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Table 5: Primary artifacts for the SFT and probe results. Probe edits use greedy decoding and fixed edit seed `Random(0)` unless otherwise noted; result-store rows also carry per-run fingerprints over split, method, decoding, seed, prompt, and code version.

Result family	Result JSON / store	Script and seed	Checkpoint or input path
Global result store	data/distill/results/results.jsonl, SHA 31ef62ccb77a	scripts/regen_tables.py; stored run seed/fingerprint	append-only store used by tables and map figures
ChartQA LoRA SFT accuracy	data/distill/poc/lora_8b_chartqa/eval_n400.json, SHA 2ce4d9175235; data/distill/poc/lora_32b_chartqa/eval_n400.json, SHA 92e59e633a71	scripts/poc_sft.py; scripts/poc_sft_32b_qlora.py; scripts/eval_sft_n400.py	train data/distill/poc/chartqa_cot_train.jsonl; adapters data/distill/poc/lora_8b_chartqa, data/distill/poc/lora_32b_chartqa/epoch_1
TabMWP LoRA SFT accuracy	data/distill/poc/eval_sft_8b_tabmwp_n400.json, SHA 95aaba7b47df; data/distill/poc/eval_sft_32b_tabmwp_n400.json, SHA b08953be8f1f	scripts/run_wu3_stage2.sh; scripts/eval_sft_n400.py	train data/distill/poc/tabmwp_cot_train.jsonl; adapters data/distill/poc/lora_8b_tabmwp, data/distill/poc/lora_32b_tabmwp/epoch_4
ChartQA faithfulness battery	data/distill/poc/battery_8b_present.json, SHA 0ff0da834fc9; data/distill/poc/battery_32b_present.json, SHA 34a57e280224; masked companions in same directory	scripts/battery_n400.py, edit seed 0	same ChartQA adapters; test split data/distill/chartqa/test_cases_400.jsonl
P0-4 local numeric control	data/distill/poc/battery_p0_local_chartqa8b_present.json, SHA e8f8867af47f	scripts/battery_n400.py, SHA 63b51bf0af0a; interventions corrupt local_num, edit seed 0	ChartQA 8B adapter; test split data/distill/chartqa/test_cases_400.jsonl
P0-4 semantic counterfactual diagnostic	data/distill/poc/battery_p0_semantic_chartqa8b_present.json, SHA 9784029ee5a2; rewrite cache data/distill/poc/semantic_cf_p0_chartqa8b_present.jsonl, SHA 1d54f6ee7890	scripts/battery_n400.py, SHA 63b51bf0af0a; interventions corrupt semantic_cf, edit seed 0; ORCH rewrite endpoint with explicit ChartQA 8B consent	same ChartQA adapter; test split data/distill/chartqa/test_cases_400.jsonl
P0-5 cross-family replication	data/distill/poc/battery_p0_cross_family_internvl35_chartqa_present.json, SHA 3403bd84b51d; base/probe caches data/distill/poc/internvl35_chartqa_base_cot.jsonl, SHA 017cdeddd221; data/distill/poc/internvl35_chartqa_probe.jsonl, SHA 1ddb3f08b58	scripts/battery_cross_family_endpoint.py, SHA cbeb6e748946; interventions corrupt shuffle, edit seed 0; endpoint greedy decode	InternVL3.5-8B served locally; ChartQA test split data/distill/chartqa/test_cases_400.jsonl
P3-1 generation-time (in-place) probe	data/distill/poc/battery_p3_online_chartqa8b_present.json, SHA e21d22b46ff9; no-visual companion data/distill/poc/battery_p3_online_chartqa8b_maskedB.json, SHA d8fce3d310e2; matched two-pass data/distill/poc/battery_p3_twopass_chartqa8b_present.json, SHA 083ccf248e8b	scripts/battery_online_p3.py, SHA b82ac920830c, greedy, edit seed 0; pre-registration docs/preregistration_p3_inplace.md	retrained ChartQA 8B adapter data/distill/poc/lora_8b_chartqa; test split data/distill/chartqa/test_cases_400.jsonl
TabMWP faithfulness battery	data/distill/poc/battery_tabmwp8b_present.json, SHA c27d11005883; data/distill/poc/battery_tabmwp32b_present.json, SHA 078a37c4e4c6; masked companions in same directory	scripts/battery_n400.py, edit seed 0	same TabMWP adapters; test split data/distill/tabmwp/test_cases_400.jsonl
Faithfulness statistical and selection audit	data/distill/results/faithfulness_stats.json, SHA 28743bfa46e7; Markdown summary data/distill/results/faithfulness_stats.md, SHA fd789992538a	scripts/faithfulness_stats.py, SHA 5f4318e19dc8; CPU-only fixed bootstrap seeds	consumes ChartQA/TabMWP battery files, local/semantic-control and cross-family battery files when present, SFT prediction JSONL files, and FinQA targeted battery files; emits raw-to-probe flow, answer-eval strata, gain-subset, local/semantic/P0-5 tests, and sensitivity bounds
FinQA targeted curriculum probe	data/distill/poc/battery_n1_8b.json, SHA 03dfe86ad4e; data/distill/poc/battery_n1_32b.json, SHA c808008b06b9; text-teacher and dense-control companions in same directory	scripts/build_n1_curriculum.py; scripts/run_n1_sft.sh; scripts/battery_n1_targeted.py, edit seed 0	train data/distill/finqa/curriculum_dev.*; probe split data/distill/finqa/curriculum_test_strict.jsonl; LoRA and QLoRA adapters under data/distill/poc/lora_*.finqa.*
Dense 8B control	data/distill/poc/full_8b_chartqa/full_sft_summary.json; data/distill/poc/full_8b_tabmwp/full_sft_summary.json; dense battery files under data/distill/poc/battery_full8b_*	scripts/audit_full_sft_8b_nonvideo.py; scripts/eval_full_sft_n400.py; scripts/export_full_sft_8b_nonvideo_evidence.py	full-SFT checkpoints under data/distill/poc/full_8b_*; summaries record trainable fraction and freeze policy